Electric Charges and Fields CHEAT SHEET

ELECTRIC CHARGE

two kinds: like charges repel, unlike attract

rubbing moves electrons; it never makes charge

conserved: +6 nC and −2 nC touch → +2 nC each

quantised: $q = ne$, $e = 1.60 \times 10^{-19}$ C

100 nC = 6.25×10^{11} electrons; 2.4×10^{-19} C = 1.5e: impossible

conductors: free electrons; insulators: none

induction: a charge near a conductor separates its charge

CONSTANTS USED

$e = 1.60 \times 10^{-19}$ C; $m_e = 9.11 \times 10^{-31}$ kg

$k = 8.99 \times 10^9$ N m²/C²; $\varepsilon_0 = 8.85 \times 10^{-12}$ C²/N m²

air sparks at about 3×10^6 N/C (dry, sea level)

water's dipole moment 6.2×10^{-30} C m

COULOMB'S LAW AND SUPERPOSITION

$F = k\frac{q_1 q_2}{r^2}$ along the line of centres

$k = \frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^9$ N m²/C²

Coulomb 1785: 36° → 36° twist, 18° → 144°, 8½° → 576°

two 1 C charges, 1 m: 8.99×10^9 N (917,000 t)

two electrons: electric / gravity ≈ 4×10^{42}

superposition: work out each pair, add as arrows

+4 μC, +1 μC, 30 cm: no force 20 cm from the 4 μC

EDGE TESTS

r grows: F, E fade to 0, never reach it

dipole, far: $2kp/r^3$ is 2 % low at 10a, 0.2 % at 30a

closed surface, no charge inside: $\Phi = 0$ (E need not be)

finite sheet: 1 % low 1 cm away, under ⅓ at R

inside a shell: E = 0 everywhere, not only the centre

THE ELECTRIC FIELD

$E = \frac{F}{q_0}$ (a small positive test charge)

point charge: $E = k\frac{Q}{r^2}$; $F = qE$

100 nC: 89,900 N/C at 10 cm, 899 N/C at 1 m

electron in 10^4 N/C: $a = 1.76 \times 10^{15}$ m/s²

across 1 cm from rest: 3.4 ns, 5.9×10^6 m/s (sim.)

the field is there with or without a test charge

FIELD LINES

start on +, end on −; never cross

crowded lines = strong field; density falls as $1/r^2$

lines follow the charge: +2q sends 32, −q takes 16

they show the push, not the path (simulated)

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ELECTRIC FLUX

$\Phi = EA\cos\theta$ (θ: field to the area vector)

1000 N/C, 0.5 m², 60° → 250 N m²/C

closed surface: area vectors out; in counts −

THE ELECTRIC DIPOLE

$p = q \times 2a$, from − to +

axis: $E = \frac{2kpr}{(r^2 - a^2)^2} \to \frac{2kp}{r^3}$ along p

equator: $E = \frac{kp}{\sqrt{(r^2 + a^2)^3}} \to \frac{kp}{r^3}$ against p

uniform field: no net force; $\tau = pE\sin\theta$

released at 60°, it swings to −60° (simulated)

paper to a comb: tiny dipoles pulled to the stronger field

GAUSS'S LAW

$\Phi = \frac{q_inside}{\varepsilon_0}$ for ANY closed surface

sphere: $E \cdot 4\pi r^2 = q/\varepsilon_0$ at every r

lopsided box, tile by tile: q/ε_0; charge outside: 0

charge outside changes E on the surface, not Φ

symmetry gives the direction, Gauss the size

WIRE, SHEET AND SHELL

wire: $E = \frac{\lambda}{2\pi\varepsilon_0 r}$ (1/r)

$\lambda = 1\ \mu C/m$: 1.8×10^5 N/C at 10 cm

sheet: $E = \frac{\sigma}{2\varepsilon_0}$ (no r: the same near or far)

two plates $\pm\sigma$: $\frac{\sigma}{\varepsilon_0}$ between, 0 outside

shell: $k\frac{Q}{r^2}$ outside; 0 inside

10 cm metal ball in air: at most 3.3 μC

THE SMOKE PLATES

$\pm 2\ \mu C/m^2$: $E = \sigma/\varepsilon_0 = 2.26 \times 10^5$ N/C everywhere between

1 μm speck, 100 extra electrons: $F = 3.6 \times 10^{-12}$ N

about 700 × its weight; drifts 2.1 cm/s (simulated)

CHECK BEFORE YOU TRUST IT

before you trust an answer about fields:

1 does it come out in N/C (or N m²/C for flux)?

2 did you add the fields as arrows, not numbers?

3 Gauss: did you count only the charge inside?

PYTHON AND THE SIMULATIONS

Python: F = 8.99e9 * q1 * q2 / r**2

E = 8.99e9 * Q / r**2; flux = E * A * math.cos(th)

field lines: step along E/|E| (RK4), stop at a − charge

sums: a wire, a disc, a shell, piece by piece

each checked against its formula